Junkai Dong

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EMPLOYMENT

Stanford Science Fellow Sep 2026 – Present
Stanford University, Stanford, CA

Quantitative Research Intern (Return Offer Received) Jun 2026 – Aug 2026
Citadel LLC, New York, NY

EDUCATION

Doctor of Philosophy in Physics Aug 2021 – May 2026
Harvard University, Cambridge, MA
Advisor: Ashvin Vishwanath
Thesis: *Two Silhouettes of Strongly Correlated Topological Phases: Fractionalization and Crystallization*

Master of Arts in Physics Aug 2021 – May 2024
Harvard University, Cambridge, MA

Bachelor of Arts in Physics and Mathematics, *summa cum laude* Aug 2017 – May 2021
Cornell University, Ithaca, NY
Thesis: *Deformed WZW Models and Averaging in Quantum Gravity*

HONORS AND AWARDS

- Stanford Science Fellowship, Stanford University, 2026
- Thouless Fellowship, University of Washington, 2026 (Declined)
- KITP Graduate Fellow, Kavli Institute for Theoretical Physics, University of California Santa Barbara, 2024
- Gertrude and Maurice Goldhaber Prize, Harvard University, 2024
- Certificate of Special Recognition, Harvard Undergraduate Association, 2023
- Purcell Fellowship, Harvard University, 2021
- Donald R. Yennie Prize in Physics, Cornell University, 2021
- Bethe Thesis Prize in Physics, Cornell University, 2021

SKILLS

Languages Chinese (native), English (bilingual)
Programming Languages Julia, Python, C++
Software bash, slurm, OpenMP, MPI, MATHEMATICA, LaTeX

MANUSCRIPTS

First or Co-First Author

22. J. Dong*, T. Soejima*, D. E. Parker, and A. Vishwanath, "Crystals Caught Doping: Metallic Wigner Crystals in Rhombohedral Graphene", *arXiv:2604.00114*. (Accepted by *Phys. Rev. Lett.*)
21. J. Dong*, O. E. Sommer*, T. Soejima, D. E. Parker, and A. Vishwanath, "Phonons in Electron Crystals with Berry Curvature", *Proc. Natl. Acad. Sci. USA* **122**(38), e2515532122 (2025). *arXiv:2503.16390*.

20. T. Soejima*, J. Dong*, A. Vishwanath, and D. E. Parker, "λ-Jellium Model for the Anomalous Hall Crystal", *Phys. Rev. Lett.* **135**, 186505 (2025). *arXiv:2503.12704*.

19. T. Soejima*, J. Dong*, T. Wang, T. Wang, M. P. Zaletel, A. Vishwanath, and D. E. Parker, "Anomalous Hall Crystals in Rhombohedral Multilayer Graphene II: General Mechanism and a Minimal Model", *Phys. Rev. B* **110**, 205124 (2024). *arXiv:2403.05522*. (Editor's Suggestion)

18. J. Dong*, T. Wang*, T. Wang*, T. Soejima, M. P. Zaletel, A. Vishwanath, and D. E. Parker, "Anomalous Hall Crystals in Rhombohedral Multilayer Graphene I: Interaction-Driven Chern Bands and Fractional Quantum Hall States at Zero Magnetic Field", *Phys. Rev. Lett.* **133**, 206503 (2024). *arXiv:2311.05568*. (Featured in PRL's Collection of the Year 2024, Journal Club for Condensed Matter Physics, Nature News and Physics World.)

17. J. Dong, J. Wang, P. J. Ledwith, A. Vishwanath, and D. E. Parker, "Composite Fermi Liquid at Zero Magnetic Field in Twisted $MoTe_2$", *Phys. Rev. Lett.* **131**, 136502 (2023). *arXiv:2306.01719*. (Editor's Suggestion, Featured in *APS Physics*)

16. J. Dong, P. J. Ledwith, E. Khalaf, J. Y. Lee, and A. Vishwanath, "Many-Body Ground States from Decomposition of Ideal Higher Chern Bands: Applications to Chirally Twisted Graphene Multilayers", *Phys. Rev. Research* **5**, 023166 (2023). *arXiv:2210.13477*.

15. J. Dong, J. Wang, and L. Fu, "Dirac electron under periodic magnetic field: Platform for fractional Chern insulator and generalized Wigner crystal", *arXiv:2208.10516*.

14. J. Dong, T. Hartman, and Y. Jiang, "Averaging over moduli in deformed WZW models", *J. High Energ. Phys.* **2021**, 185 (2021). *arXiv:2105.12594*.

13. J. Dong, V. Juricic, and B. Roy, "Topolectric circuits: Theory and construction", *Phys. Rev. Research* **3**, 023056 (2021). *arXiv:2008.11202*.

12. J. Dong and E. Mueller, "Exact Topological Flat Bands from Continuum Landau Levels", *Phys. Rev. A* **101**, 013629 (2020). *arXiv:1910.08429*.

11. J. Dong, Y. Chen, D. Xu, and Z.-Q. Yin, "Greenberger-Horne-Zeilinger test for multi-dimension and arbitrary time nodes entangled histories", *Sci. Bull.* **62**(18), pp.1235-1238 (2016). *arXiv:1610.04296*.

Others

10. C. E. Henzinger, J. R. Ehrets, R. Fushio, J. Dong, T. Werkmeister, M. E. Wesson, K. Watanabe, T. Taniguchi, A. Vishwanath, B. I. Halperin, A. Yacoby, and P. Kim, "Controlled localization of anyons in a graphene quantum Hall interferometer", *arXiv:2603.11182*.

9. A. Valenti, Y. Vituri, Y. Yang, D. E. Parker, T. Soejima, J. Dong, M. A. Morales, A. Vishwanath, E. Berg, and S. Zhang, "Quantum Geometry Driven Crystallization: A Neural-Network Variational Monte Carlo Study", *arXiv:2512.07947*.

8. M. Uzan, W. Zhi, M. Bocarsly, J. Dong, S. Dutta, N. Auerbach, N. S. Kander, M. Labendik, Y. Myasoedov, M. E. Huber, K. Watanabe, T. Taniguchi, D. E. Parker, and E. Zeldov, "hBN alignment orientation controls moiré strength in rhombohedral graphene", *arXiv:2507.20647*. (Accepted by *Nature Physics*.)

7. T. Soejima, J. Dong, O. E. Sommer, D. E. Parker, and A. Vishwanath, "Topological constraint on crystalline current", *Phys. Rev. Lett.* **136**, 186601 (2026). *arXiv:2507.18611*.

6. R. Fan, J. Dong, and A. Vishwanath, "Simulating the non-unitary Yang-Lee conformal field theory on the fuzzy sphere", *J. High Energ. Phys.* **2026**, 41 (2026). *arXiv:2505.06342*.

5. P. J. Ledwith, J. Dong, A. Vishwanath, and E. Khalaf, "Nonlocal Moments in the Chern Bands of Twisted Bilayer Graphene", *Phys. Rev. X* **15**, 021087 (2025). *arXiv:2408.16761*.

4. Q. Li, J. Dong, P. J. Ledwith, and E. Khalaf, "Constraints on real space representations of Chern bands", *arXiv:2407.02561*.

3. M. Fujimoto, D. E. Parker, J. Dong, E. Khalaf, A. Vishwanath, and P. J. Ledwith, "Higher vortexability: zero field realization of higher Landau levels", *Phys. Rev. Lett.* **134**, 106502 (2025). *arXiv:2403.00856*. (Editor's Suggestion, Featured in PRL's Collection of the Year 2025)

2. Q. Gao, J. Dong, P. J. Ledwith, D. E. Parker, and E. Khalaf, "Untwisting moiré physics: Almost ideal bands and fractional Chern insulators in periodically strained monolayer graphene", *Phys. Rev. Lett.* **131**, 096401 (2023). *arXiv:2211.00658*. (PRL cover)

1. J. Dong, V. Elser, G. Gyawali, K. Y. Jee, J. Kent-Dobias, A. Mandaiya, M. Renz, and Y. Su, "Glass phenomenology in the hard matrix model", *J. Stat. Phys.* **2021**(9), 093302 (2021). *arXiv:1912.07558*.

PRESENTATIONS

Invited Talks

- *Electron Crystallography in Rhombohedral Graphene*, Long Ju Group Meeting, MIT, May 2026.
- *Anomalous Hall Crystal and its Phonons*, Kids Seminar, Harvard University, Feb 2026.
- *Anomalous Hall Crystal and its Phonons*, Center for Computational Quantum Physics, Flatiron Institute, Dec 2025.
- *Anomalous Hall Crystal and its Phonons*, Geballe Laboratory for Advanced Materials, Stanford University, Dec 2025.
- *Anomalous Hall Crystal and its Phonons*, Thouless Institute for Quantum Matter Seminar, University of Washington Seattle, Dec 2025.
- *Anomalous Hall Crystal and its Phonons*, MRS Spring Meeting and Exhibit, Apr 2025.
- *Anomalous Hall Crystals in Rhombohedral Multilayer Graphene*, Joonho Lee Group Meeting, Harvard University, Mar 2025.
- *Stripping off the Magnetic Field from the Lowest Landau Level*, KITP Locals Lunch, Aug 2024.
- *Anomalous Hall Crystal in Rhombohedral Multilayer Graphene*, KITP Condensed Matter Meeting, Jul 2024.
- *Anomalous Hall Crystal in Rhombohedral Multilayer Graphene*, Quantum Theory Seminar, Cornell University, Apr 2024.
- *Composite Fermions Form and Flow without a Magnetic Field*, Physical Review Journal Club, American Physical Society, Nov 2023.
- *Composite Fermi Liquid at Zero Magnetic Field in Twisted $MoTe_2$*, Special AEP Seminar, Cornell University, Aug 2023.
- *Composite Fermi Liquid at Zero Magnetic Field in Twisted $MoTe_2$*, Thouless Institute for Quantum Matter Seminar, University of Washington Seattle, Jun 2023.
- *Exact Many-Body Ground States from Decomposition of Ideal Higher Chern Bands: Applications to Chirally Twisted Graphene Multilayers*, Quantum Matter in Mathematics and Physics, Center of Mathematical Sciences and Applications, Harvard University, MA, Dec 2022.

Posters

- *Anomalous Hall Crystals in Rhombohedral Multilayer Graphene*, Thouless Institute for Quantum Matter Winter Workshop, WA, Jan 2024.
- *Composite Fermi Liquid at Zero Magnetic Field in Twisted $MoTe_2$*, Thouless Institute for Quantum Matter Winter Workshop, WA, Jan 2024.
- *Anomalous Hall Crystals in Rhombohedral Multilayer Graphene*, National High Magnetic Field Laboratory Theory Winter School, FL, Jan 2024.

- *Composite Fermi Liquid at Zero Magnetic Field in Twisted $MoTe_2$*, National High Magnetic Field Laboratory Theory Winter School, FL, Jan 2024.
- *Composite Fermi Liquid at Zero Magnetic Field in Twisted $MoTe_2$*, Princeton Summer School on Condensed Matter Physics 2023, Princeton University, July 2023.
- *Exact Many-Body Ground States from Decomposition of Ideal Higher Chern Bands: Applications to Chirally Twisted Graphene Multilayers*, Spring 2023 meeting of the Simons Collaboration on Ultra-Quantum Matter, CU Boulder, CO, May 2023.
- *Exact Many-Body Ground States from Decomposition of Ideal Higher Chern Bands: Applications to Chirally Twisted Graphene Multilayers*, National High Magnetic Field Laboratory Theory Winter School, FL, Jan 2023.
- *Exact Topological Flat Bands from Continuum Landau Levels*, ARO/AFOSR MURI Program Review Meeting, UMass Amherst, MA, Oct 2019.

TEACHING EXPERIENCE

Teaching Fellow, Fall 2025
Led discussion sections, held office hours, supervised presentations, created and graded homework, and took lecture notes for PHYSICS 268R (Quantum Phases of Matter).
Teaching Fellow, Spring 2025
Led office hours, supervised presentations, and gave lectures for PHYSICS 195B (Introduction to Quantum Materials and Devices).
Teaching Fellow, Fall 2023
Held sections, led office hours, and graded homework and exams for PHYSICS 195A (Introduction to Solid State Physics).
Teaching Assistant, Fall 2019
Held one-hour study halls for PHYS 7651 (Quantum Field Theory 1) every week.
Teaching Assistant, Fall 2020
Graded homework and answers questions online for PHYS 7681 (Quantum Information Processing).

SERVICE

APS March Meeting Session Chair, 2025
Chaired session J27: Emergent Phases and Transport Phenomena in Twisted and Multilayer Graphene Systems.
APS March Meeting Session Chair, 2024
Chaired session Z07: Magnetic Topological Semimetals III.
Journal Referee, 2023-Present
Provided peer review for:

- Nature ($\times 3$)
- Nature Physics ($\times 2$)
- Physical Review X ($\times 10$)
- Physical Review Letters ($\times 8$)
- Physical Review Research ($\times 3$)
- Physical Review B ($\times 32$)
- Journal of Physics: Condensed Matter ($\times 3$)
- Reports on Progress in Physics: Original Research ($\times 1$)

Cornell Alumni Admissions Ambassadors Network Volunteer, 2024
Met with prospective applicants to discuss details about undergraduate experience at Cornell.
Harvard Organ Society Recital Coordinator, 2022-2024

Revived the Busch Midday Recital Series after the COVID pandemic. Invited and hosted professional organists. Doubled audience size during tenure.